Criterion 8: Program Outcomes (POs) and Assessment

Department of Computer Science and Engineering

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8.1: Program outcome (PO) statements and their mapping with the PEOs

Graduates of the B.Sc. in Computer Science and Engineering (CSE) program are expected to achieve the following listed Program Outcomes (POs) by the time of graduation.

Program Outcomes

POs	Descriptions
PO1	Apply knowledge of mathematics, natural science, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to the solution of complex engineering problems.
PO2	Identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences. (WK1 to WK4)
PO3	Design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations. (WK5)
PO4	Conduct investigations of complex problems using research-based knowledge (WK8) and research methods including design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions.

Continued

POs	Descriptions
PO5	Create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems, with an under standing of the limitations. (WK6)
PO6	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice and solutions to complex engineering problems. (WK7)
PO7	Understand and evaluate the sustainability and impact of professional engineering work in the solution of complex engineering problems in societal and environmental contexts. (WK7)
PO8	Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice. (WK7)
PO9	Function effectively as an individual, and as a member or leader in diverse teams and in multi-disciplinary settings.

Continued

POs	Descriptions
PO10	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Demonstrate knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

8.2: How course outcomes are developed and reviewed

1. **Instructors draft the course package:** outline, contents, assessment plan, and CO–PO mappings using the approved OBE curriculum and syllabus.
2. **Course coordinators harmonize sections:** align contents, assessment, and mappings across multiple sections of a course.
3. **The Departmental Committee reviews:** examine proposed materials, previous attainment, CQI findings, and instructor, student, alumni, and employer feedback.
4. **The Syllabus Committee examines curricular changes:** check alignment with the mission, POs, and PEOs; significant changes follow institutional approval through the Academic Council.
5. **The department finalizes the COs:** incorporate approved improvements into outcomes, outlines, and assessment strategies.

The recurring work

Use evidence and feedback from earlier offerings to improve the next course design.

Source: Criterion 8 SAR, Section 8.2, p. 127.

8.2.1: What is documented for each course outcome?

The CO specification

- A specific, measurable statement of expected knowledge, skills, or competence.
- One or more mapped POs.
- Learning-domain codes.
- Delivery methods and learning activities.
- Assessment tools that can demonstrate the outcome.

Example: AI theory, CO3

Student work: recommend solutions to real-life AI problems using existing techniques.

Alignment: PO3 (design).
Domain: C6.
Delivery: lecture and demonstration.
Evidence: mid-term and final examinations.

Domain families: C = cognitive; A = affective; P = psychomotor. Numeric codes on the following slides are reproduced from Table 8.3.

Source: Criterion 8 SAR, Section 8.2.1 and Table 8.3, pp. 127–130.

HUM 4441: Engineering Ethics

Delivery: Lecture, group discussion, and problem solving.
Assessment: Mid-term and final examinations.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Understand moral and ethical principles governing engineering practice and decisions.	PO6	C2
CO2	Examine social and cultural implications, professional responsibility, and accountability.	PO8	C4
CO3	Develop a framework for ethical analysis and engineering decision-making.	PO8	C3
CO4	Analyze and apply ethical implications of data technologies, privacy, security, and ownership.	PO6	C4
CO5	Understand how data-driven technologies create or perpetuate bias and discrimination.	PO6	C2

Source: Criterion 8 SAR, Table 8.3, p. 128; CO descriptions summarized.

CSE 4513: Software Engineering and OOD

Delivery: Lecture and problem solving.
Assessment: Mid-term and final examinations.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Explain software development fundamentals and requirement engineering.	PO1, PO2	C2
CO2	Apply design knowledge to model large software systems.	PO2	C3
CO3	Use techniques and tools to release and maintain quality software.	PO1, PO2	C3

Work to explain

Students move from understanding the development process to modeling systems and applying software release and maintenance practices.

Source: Criterion 8 SAR, Table 8.3, p. 128; CO descriptions summarized.

Delivery: Lecture and demonstration.
Assessment: Mid-term and final examinations.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Apply general AI techniques to common problem types.	PO1	C3
CO2	Formulate problem scenarios as environments and agents, with justification.	PO2	C5
CO3	Recommend solutions to real-life AI problems using existing techniques.	PO3	C6

Work to explain

Apply AI knowledge $\longrightarrow$ formulate the problem $\longrightarrow$ propose a suitable solution.

Source: Criterion 8 SAR, Table 8.3, p. 128; CO descriptions summarized.

CSE 4712: Artificial Intelligence Lab

Delivery: Group discussion, demonstration, and problem solving.
Assessment: Lab viva, lab tasks, and lab reports.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Implement and analyze algorithms for complex navigation and strategy problems.	PO5	C3, C4
CO2	Formulate complex real-world scenarios as specific AI paradigms.	PO2	C5
CO3	Design, test, and evaluate intelligent agents under uncertainty using modern tools and frameworks.	PO4	C6, P5
CO4	Communicate analysis, design choices, and performance through structured technical reports.	PO10	P7, C5

The PO mappings here follow Table 8.3; Table 8.4 differs for CO1 and CO3 (see the review slide).

Source: Criterion 8 SAR, Table 8.3, pp. 128–129; CO descriptions summarized.

CSE 4610: Design Project

Delivery: Group discussion, demonstration, and project work.
Assessment: Presentation and project.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Identify research problems and implement innovative solutions with current technologies, tools, and algorithms.	PO1–PO5	C4
CO2	Demonstrate teamwork, ethics, responsibility, and project management to deliver a software system within constraints.	PO8, PO9, PO11	C3, A4
CO3	Design and evaluate computing solutions considering ethical, societal, and environmental implications.	PO6, PO7, PO8	C5, A3
CO4	Communicate outcomes through a structured research report and formal presentation.	PO10	C3, A3
CO5	Build a foundation for future research in the same or related domains.	PO12	C2, A2

Source: Criterion 8 SAR, Table 8.3, p. 129; CO descriptions summarized.

CSE 4700: Project/Thesis

Delivery: Discussion, demonstration, problem solving, and report writing.
Assessment: Presentation and report.

CO	Expected student work (summarized)	Mapped POs	Domains
CO1	Identify and formulate a significant research problem through literature review and analysis of existing solutions.	PO1, PO2, PO4	C4, A4
CO2	Propose methodologies, models, or systems for the identified problem.	PO3	C6, A5, P5
CO3	Conduct experiments, simulations, or data-driven investigations; evaluate solutions and interpret results.	PO4, PO5	C4, A4, P4
CO5	Prepare a technical report and present findings for technical and non-technical audiences.	PO10	C5, P5, A7
CO6	Work independently and collaboratively, apply project management, and pursue continuous learning.	PO9, PO11, PO12	C6, P5, A7

CO numbering follows the source: Table 8.3 does not list a CO4 for this course.

Source: Criterion 8 SAR, Table 8.3, pp. 129–130; CO descriptions summarized.

CSE 4800: Project/Thesis

Delivery: Discussion, demonstration, problem solving, and report writing.
Assessment: Presentation and report.

CO	Expected student work (summarized)	Mapped POs	Domains
CO2	Propose methodologies, models, or systems for the identified problem.	PO3	C6, A5, P5
CO3	Conduct experiments, simulations, or data-driven investigations; evaluate solutions and interpret results.	PO4, PO5	C4, A4, P4
CO4	Demonstrate ethical responsibility and professionalism in research, considering society and the environment.	PO6, PO7, PO8	C6, P6
CO5	Prepare a technical report and present findings for technical and non-technical audiences.	PO10	C5, P5, A7
CO6	Work independently and collaboratively, apply project management, and pursue continuous learning.	PO9, PO11, PO12	C6, P5, A7

CO numbering follows the source: Table 8.3 does not list a CO1 for this course.

Source: Criterion 8 SAR, Table 8.3, p. 130; CO descriptions summarized.

8.2.2: How the courses collectively contribute to POs

Course	PO coverage derived from the CO statements in Table 8.3
HUM 4441: Engineering Ethics	PO6, PO8
CSE 4513: Software Engineering and OOD	PO1, PO2
CSE 4711: Artificial Intelligence	PO1, PO2, PO3
CSE 4712: Artificial Intelligence Lab	PO2, PO4, PO5, PO10
CSE 4610: Design Project	PO1–PO12 (all twelve)
CSE 4700: Project/Thesis	PO1, PO2, PO3, PO4, PO5, PO9, PO10, PO11, PO12
CSE 4800: Project/Thesis	PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12

Why the distribution matters

Theory develops knowledge and analysis; laboratory work adds implementation and investigation; projects integrate technical and professional capabilities.

Source: Criterion 8 SAR, Section 8.2.2 and Tables 8.3–8.4, pp. 128–131; coverage uses Table 8.3 where the tables differ.

Source review: reconcile the CO–PO tables

Table 8.3 and Table 8.4 contain differences that should be checked before finalizing the SAR or presentation.

Course / CO	Table 8.3: statements	Table 8.4: mapping matrix
CSE 4712 / CO1	PO5	PO4
CSE 4712 / CO3	PO4	PO3
CSE 4700 / CO5	PO10	PO6
CSE 4800 / CO5	PO10	No PO marked

Treatment in this draft

Course slides and the coverage summary consistently use Table 8.3. Retain the original CO numbering and learning-domain codes pending confirmation against the approved course outlines.

Source: Criterion 8 SAR, Table 8.3, pp. 128–130; Table 8.4, p. 131.

What Sections 8.1 and 8.2 document

1. **Graduate expectations:** twelve defined POs, with an approval reference.
2. **Program alignment:** PO–PEO mappings and a rationale for every relationship.
3. **Course design:** CO statements, learning domains, delivery methods, and assessment tools for seven selected core courses.
4. **Review responsibilities:** instructor preparation, coordinator harmonization, departmental review, and formal approval of significant changes.
5. **Traceability:** each course outcome connects student work to one or more graduate outcomes.

Presentation takeaway

These sections explain how the program defines its outcomes and organizes course work to support them.

Source: Criterion 8 SAR, Sections 8.1–8.2, pp. 125–131.